

Sentinel-A2A

Runtime Security Firewall for Autonomous AI Agents

Zero-Trust security layer inspecting Model Context Protocol (MCP) tool execution and inter-agent communication payloads in real time.

PROBLEM STATEMENT

As multi-agent ecosystems and Model Context Protocol (MCP) integrations proliferate, Agent-to-Agent (A2A) communications lack real-time runtime security layers. Unmonitored payload exchanges expose AI agents to prompt injections, indirect attacks, unauthorized tool execution, and privilege escalation without an active firewall boundary.

CORE SOLUTION

Sentinel-A2A introduces a Zero-Trust runtime firewall for autonomous AI agents. It intercepts inter-agent communications and MCP tool executions in real time, leveraging **Google Gemini** for AI threat intelligence, **Google Cloud Firestore** for persistent audit logging, and a custom non-saturating risk scoring engine to evaluate, block, or allow payloads dynamically.

Google Cloud Infrastructure & Services

- **Google Gemini API:** Acts as the primary AI Security Intelligence engine, performing deep contextual payload inspection, identifying threat vectors (e.g., prompt injections, exfiltration), and rendering instant mitigation verdicts.
- **Google Cloud Firestore:** Serves as the central persistent database for storing structured security audit logs, event telemetry, risk scores, and historical execution results.
- **Google Cloud Run & GCP Ecosystem:** Provides scalable cloud infrastructure foundation for backend runtime execution, security state management, and real-time event streaming.

Replit Platform Integration (Partner Track)

- **Live App Previewing & Hosting:** Replit was utilized as the primary cloud hosting and development environment to run the full-stack application (Python/Streamlit backend exposed on port `8080`), enabling instant web-based dashboard previews.
- **Demonstration & Video Production:** Served as the live environment to run attack simulations (benign transactions vs. malicious injections), verify local in-memory fallback modes, and record real-time security telemetry for the official 3-minute video showcase.

Developer Contributions & Team Architecture

Aditya Srivastava

FOUNDER

FULL-STACK & CYBERSEC

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Designed and built the entire end-to-end full-stack architecture—including the Streamlit frontend, local fallback state machine, and backend inspection engine. Studied cybersecurity principles, inter-agent attack vectors, and MCP tool safety to integrate Google Gemini threat intelligence and Firestore audit logging into a cohesive runtime firewall.

Niket Jha

PARTNER

MECHANICAL & HARDWARE LEAD

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Leads the physical prototype hardware expansion and mechanical system integration, bridging software firewall policies with real-world autonomous vehicle hardware controls.

Future Prospects & Mechanical Expansion

Physical Autonomous Vehicle Prototype (Managed by Partner Niket Jha):

- **Hardware Testbed Platform (Budget: ₹5,000 – ₹7,000):** Engineering a low-cost, scaled autonomous rover equipped with microcontrollers (ESP32/Raspberry Pi) and motor drivers to physicalize agent control interactions.
- **Real-World Intersection Simulation:** Deploying two independent self-driving vehicle agents approaching a physical intersection, where movement coordination payloads are routed directly through Sentinel-A2A.
- **Hardware-Level Interception & Safety Cut-off:** Testing Sentinel-A2A's capability to issue instant physical brake triggers or emergency kill-switches when an agent receives corrupted or malicious directional commands over the hardware proxy.

Software & Gateway Expansion:

- **Sub-Millisecond Edge Proxies:** Deploy lightweight Rust/C++ middleware embedded directly onto vehicular CAN buses or ROS2 robotics nodes.
- **Multi-LLM Security Consensus:** Combine evaluations from Gemini and specialized local security models for weighted voting on high-risk physical actions.